

MTH166

Lecture-1

**Basics of Ordinary Differential
Equations**

Unit 1: Ordinary Differential Equations

(Book: Advanced Engineering Mathematics by Jain and Iyengar, Chapter-4)

Topic:

Separable form (ODE)

Learning Outcomes:

1. Definition of Ordinary Differential Equation.
2. Solution of ODE by Variable separation method
3. First order Homogeneous Differential Equations.

Ordinary Differential Equation (ODE):

A *differential equation* can be defined as an equation containing derivatives of various orders and the variables.

A differential equation which involves only one independent variable is called an *ordinary differential equation* (ODE).

Let y be the dependent variable and x be an independent variable.

We denote the derivatives as:

$$\frac{dy}{dx} = y', \frac{d^2y}{dx^2} = y'', \frac{d^3y}{dx^3} = y''', \text{ etc.}$$

Ordinary Differential Equation (ODE):

Order: The order of a differential equation is the order of the highest order derivative occurring in the equation.

Degree: The degree of a differential equation is the degree (power) of the highest order derivative occurring in the equation, after the equation has been made free of radicals and functions in its derivatives.

Linear: A differential equation is linear, when the dependent variable and its derivatives occur only in first degree and no products of the dependent variable and its derivatives or of various order derivatives occur.

Non-Linear: A differential equation which is not linear, is called non-linear.

Ordinary Differential Equation (ODE):

Some examples of ordinary differential equations are:

1. $y' = 6x^2$

Order = 1, Degree = 1, Linear.

2. $y'' + 16y = 2x$

Order = 2, Degree = 1, Linear.

3. $x^2y'' - xy' + 6y = \log x$

Order = 2, Degree = 1, Linear.

4. $y'y'' + y^2 = x^2$

Order = 2, Degree = 1, Non-Linear.

Polling Quiz

The equation: $yy' + y - x^2 = 0$ is a:

- (A) A linear ODE with order 1 and degree 1
- (B) A non-linear ODE with order 1 and degree 1
- (C) A linear ODE with order 1 and degree 2
- (D) A non-linear ODE with order 2 and degree 2

ODE Solvable by Variable Separable form

Let the given ODE be:

$$y' = f(x, y) \quad \text{where } f(x, y) = g(x)h(y)$$

$$\Rightarrow \frac{dy}{dx} = g(x)h(y)$$

$$\Rightarrow \frac{dy}{h(y)} = g(x)dx \quad (\text{Separating the variables})$$

$$\Rightarrow \int \frac{dy}{h(y)} = \int g(x)dx \quad (\text{Integrating both sides})$$

$$\Rightarrow A(y) = B(x) + C \quad \text{which is the required solution.}$$

Solve the following differential equations:

Problem 1: $(x \log x)y' = y$

Solution: $(x \log x) \frac{dy}{dx} = y \quad (1)$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{(x \log x)} \quad (\text{Separating the variables})$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{\frac{1}{x} dx}{\log x} \quad (\text{Integrating both sides})$$

$$\Rightarrow \log y = \log(\log x) + \log c \quad \left(\int \frac{f'(x)}{f(x)} dx = \log f(x) \right)$$

$$\Rightarrow \log y = \log(c \log x)$$

$$\Rightarrow y = c \log x \quad \textbf{Answer.}$$

Problem 2: $y' = e^{x+y} + x^2 e^y$

Solution: $\frac{dy}{dx} = e^{x+y} + x^2 e^y = e^y(e^x + x^2)$ (1)

$$\Rightarrow \frac{dy}{e^y} = (e^x + x^2)dx \quad (\text{Separating the variables})$$

$$\Rightarrow \int e^{-y} dy = \int (e^x + x^2) dx \quad (\text{Integrating both sides})$$

$$\Rightarrow -e^{-y} = e^x + \frac{x^3}{3} + c$$

$$\Rightarrow e^x + e^{-y} + \frac{x^3}{3} + c = 0 \quad \textbf{Answer.}$$

Problem 3: $axy' = by, a \neq 0$

Solution: $ax \frac{dy}{dx} = by \quad (1)$

$$\Rightarrow \frac{dy}{y} = \frac{b}{a} \frac{dx}{x} \quad (\text{Separating the variables})$$

$$\Rightarrow \int \frac{dy}{y} = \frac{b}{a} \int \frac{dx}{x} \quad (\text{Integrating both sides})$$

$$\Rightarrow \log y = \frac{b}{a} \log x + \log c$$

$$\Rightarrow \log y = \log \left(cx^{b/a} \right)$$

$$\Rightarrow y = \left(cx^{b/a} \right) \quad \textbf{Answer.}$$

First order Homogeneous ODE

A first order ODE: $y' = f(x, y)$ is said to be a homogeneous equation, if $f(x, y)$ is a homogeneous function, that is, ODE is of the form:

1. $y' = \frac{y}{x}$ (Homogeneous function)

2. $y' = \frac{x^2 - y^2}{xy}$ (Homogeneous function)

3. $y' = \frac{x^3 + y^3 + x^2 y}{x^3 - y^3}$ (Homogeneous function)

The equation: $y' = \frac{x^3 - y^3 - y^2x}{x^3 + y^3}$ is Homogeneous:

(A) Yes (Answer)

(B) No

